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Attention-Aided MMSE for OFDM Channel Estimation

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***Abstract*—This paper proposes an Attention-aided MMSE (A-MMSE) framework for OFDM channel estimation. In contrast to conventional deep learning methods that typically operate as direct Neural Estimators, the proposed approach adopts a Neural Filtering strategy. We employ a Two-Stage Attention Encoder to extract channel correlations and generate adaptive linear filter coefficients. By explicitly modeling the separable frequency and temporal dependencies, the network effectively captures non-stationary channel statistics that are challenging for conventional methods. This structure allows the inference phase to utilize efficient linear matrix-vector multiplication. Simulation results indicate that the A-MMSE provides improved NMSE and BER performance compared to existing baselines while reducing computational complexity.**

***Index Terms*—Orthogonal frequency-division multiplexing, channel estimation, minimum mean-squared error, Transformer attention, deep learning, 5G/6G networks.**

I. Introduction

Orthogonal frequency division multiplexing (OFDM) stands as a cornerstone waveform for modern wireless communication systems, underpinning 5G New Radio (NR) and paving the way for future 6G technologies [1]–[3]. A fundamental challenge in realizing the full potential of OFDM systems is channel estimation, which is essential for coherent detection and symbol recovery. The goal is to accurately recover the channel frequency response from sparse pilot symbols corrupted by noise and interference.

For decades, this problem has been addressed primarily through signal processing (SP)-based approaches [4]. The least square (LS) estimator is widely used due to its simplicity but suffers significantly from noise enhancement. The minimum mean-squared error (MMSE) estimator, while statistically optimal for Gaussian channels, relies heavily on accurate knowledge of the channel's second-order statistics, such as the covariance matrix. In practical scenarios—especially those involving high mobility or non-stationary environments—acquiring these statistics is non-trivial, and the requisite matrix inversions impose a prohibitive computational burden.

To overcome the limitations of model-based SP methods, deep learning has emerged as a powerful data-driven alternative. Recent works, such as ChannelNet [5] and Channelformer [6], leverage the universal approximation capability of deep neural networks (DNNs) to learn channel distributions directly from data. However, the majority of these existing approaches function as *Neural Estimators*. In this paradigm, a deep neural network acts as an end-to-end black box that directly maps noisy pilot observations to channel estimates. While effective in capturing non-linear channel imperfections, this architecture inherently couples the learning capability with the inference process. Consequently, it necessitates propagating input signals through deep layers of computationally expensive non-linear operations (e.g., convolutions, self-attention) for every single channel estimation instance. This high inference complexity and lack of interpretability create a significant bottleneck for real-time implementation in resource-constrained mobile devices.

In this paper, we propose a paradigm shift from Neural Estimation to *Neural Filtering*. We introduce the Attention-aided MMSE (A-MMSE), a hybrid framework that harmonizes the adaptability of deep learning with the structural efficiency of linear signal processing. The core idea is to use a DNN not to perform the estimation, but to design the estimator. Specifically, our model learns to generate the optimal linear MMSE filter coefficients tailored to the current channel context using a novel Two-Stage Attention Encoder. This encoder is explicitly designed to capture the separable frequency and temporal correlation structures of OFDM channels.

By decoupling the filter generation from the channel estimation process, the A-MMSE enables the inference phase to rely on linear matrix-vector multiplication. This Neural Filtering approach mitigates the computational burden associated with non-linear operations in conventional neural estimators, while also addressing the practical challenges of statistical mismatch inherent in classical MMSE. Simulation results confirm that the proposed method achieves improved estimation accuracy with reduced computational complexity compared to existing baselines, highlighting its potential for efficient deployment in next-generation wireless systems.

The remainder of this paper is organized as follows. Section II describes the system model. Section III reviews conventional channel estimation approaches, including both SP-based and DNN-based methods. Section IV details the proposed A-MMSE framework, focusing on its neural filtering mechanism and the two-stage attention encoder. Section V presents the simulation results regarding estimation accuracy and computational complexity, followed by the conclusion in

Section VI.

Notations: We denote column-wise vectorization by $vec(\cdot)$. $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of a complex quantity, respectively. $||\cdot||_F$ denotes the Frobenius norm, and $\mathbb{E}[\cdot]$ stands for statistical expectation. We use $\circ$ for the Hadamard product, $\otimes$ for the Kronecker product, and $[\cdot;\cdot]$ for vertical concatenation.

II. System Model

We consider a single OFDM slot consisting of M OFDM symbols and N subcarriers, totaling NM resource elements. Let $H[n, m]$ denote the channel frequency response at the n^{th} subcarrier and m^{th} symbol. The received signal $Y[n, m]$ is given by

$$Y[n, m] = H[n, m]X[n, m] + Z[n, m], \quad (n, m) \in \mathcal{T} \tag{1}$$

where $X[n, m]$ is the transmitted signal, $Z[n, m]$ is additive Gaussian noise, and $\mathcal{T}$ is the set of all resource indicies. We denote the set of pilot positions are as $\mathcal{P}$, where $X[n, m]$ is known to the receiver for $(n, m) \in \mathcal{P}$. The received signal model can be vectorized as

$$\mathbf{Y} = \mathbf{H} \circ \mathbf{X} + \mathbf{Z} \tag{2}$$

where $\circ$ denotes the Hadamard(element-wise) product. The objective is to estimate the full channel matrix $\mathbf{H}$ given the pilot observations $\mathbf{Y}_p$ extracted from $\mathcal{P}$.

III. Conventional Method

Existing OFDM channel estimation techniques can be broadly categorized into two primary paradigms: classical Signal Processing (SP)-based approaches and modern Deep Neural Network (DNN)-based approaches. The SP-based methods typically rely on rigorous mathematical formulations and estimation theory, often utilizing prior knowledge of channel statistics such as noise variance and correlation matrices to minimize estimation errors. Conversely, DNN-based approaches adopt a data-driven strategy, utilizing neural networks to learn the nonlinear mapping from noisy pilot observations to channel estimates without requiring explicit statistical modeling. In this section, we briefly review representative methods from both categories—specifically LS and MMSE for SP, and ChannelNet and Channelformer for DNN—to highlight their respective limitations in terms of statistical dependence and computational complexity.

A. SP-based Approaches

Conventional signal processing (SP)-based channel estimation techniques generally fall into two categories depending on their use of channel statistics.

LS: The least square (LS) method [4], [7] represents the most intuitive approach, estimating the channel by performing element-wise division of the received pilot signals by the transmitted pilot sequence. This is mathematically expressed as $\hat{H}_{LS} = Y_p/X_p$. The LS method offers the advantage of low implementation complexity as it operates without prior knowledge of channel statistics. However, it suffers from significant performance degradation, particularly in low signal-to-noise ratio (SNR) regimes, because it directly amplifies the noise component inherent in the received signal.

MMSE: The Linear MMSE (MMSE) [8] addresses the noise enhancement issue by minimizing the mean squared error (MSE). Assuming a Gaussian channel environment, the linear MMSE (LMMSE) estimator provides the statistically optimal solution and is given by [9]:

$$\begin{aligned} \text{vec}(\hat{\mathbf{H}}_{\text{MMSE}}) &= \mathbb{E}\left[\mathbf{H} \mid Y[n, m], (n, m) \in \mathcal{P}\right] \\ &= \mathbf{R}_{\text{vec}(\mathbf{H})\mathbf{H}_p} \mathbf{R}_{\mathbf{Y}_p\mathbf{Y}_p}^{-1} \mathbf{Y}_p, \end{aligned} \tag{3}$$

where $\text{vec}(\hat{\mathbf{H}}_{\text{MMSE}}) \in \mathbb{C}^{NM}$ denotes the vectorized full channel matrix, and $\mathbf{H}_p \in \mathbb{C}^L$ corresponds to the vectorized frequency response at the pilot positions. Here, $\mathbf{R}_{\text{vec}(\mathbf{H})\mathbf{H}_p} \in \mathbb{C}^{NM \times L}$ is the cross-covariance matrix between $\text{vec}(\mathbf{H})$ and $\mathbf{H}_p$, whereas $\mathbf{R}_{\mathbf{Y}_p\mathbf{Y}_p} \in \mathbb{C}^{L \times L}$ denotes the auto-covariance matrix of $\mathbf{Y}_p$.

Specifically, assuming that the noise term $\mathbf{Z}$ is independent and identically distributed (i.i.d.) Gaussian with variance σ^2, (3) can be expanded as

$$\text{vec}(\hat{\mathbf{H}}_{\text{MMSE}}) = \underbrace{\mathbf{R}_{\text{vec}(\mathbf{H})\mathbf{H}_p} \left(\mathbf{R}_{\mathbf{H}_p\mathbf{H}_p} + \sigma^2\mathbf{I}\right)^{-1}}_{\mathbf{W}_{\text{MMSE}} \in \mathbb{C}^{NM \times L}} \mathbf{Y}_p, \tag{4}$$

where $\mathbf{W}_{\text{MMSE}}$ is defined as the MMSE filter matrix. Although optimal for joint Gaussian channels, calculating the inverse of the covariance matrix is computationally prohibitive, and the performance degrades heavily when the assumed statistics mismatch the true channel conditions.

B. DNN-based Approaches

DNN-based approaches have emerged as data-driven alternatives, functioning as non-linear denoisers without requiring explicit channel statistics. These methods generally learn a direct mapping from noisy pilot observations to channel estimates. ChannelNet [5] treats the time-frequency channel response as a image. It applies super-resolution and denoising convolutional networks (CNNs) to refine coarse initial estimates derived from pilot interpolation. Channelformer [6], in contrast, adopts the Attention Transformer architecture. It employs a Multi-Head Self-Attention (MHA) mechanism to capture long-range dependencies among sparse pilots, effectively modeling global channel correlations.

C. Limitations and Motivation

Despite their respective strengths, existing approaches face fundamental limitations. First, SP-based methods like MMSE are theoretically optimal only when perfect channel statistics are available. In practice, particularly in non-stationary environments where the channel statistics vary over time, the estimator must rely on an estimated covariance $\tilde{\mathbf{R}}$ rather than the true covariance $\mathbf{R}$. This leads to a covariance-mismatched filter $\mathbf{W}_{mis}$, which incurs an unavoidable performance penalty

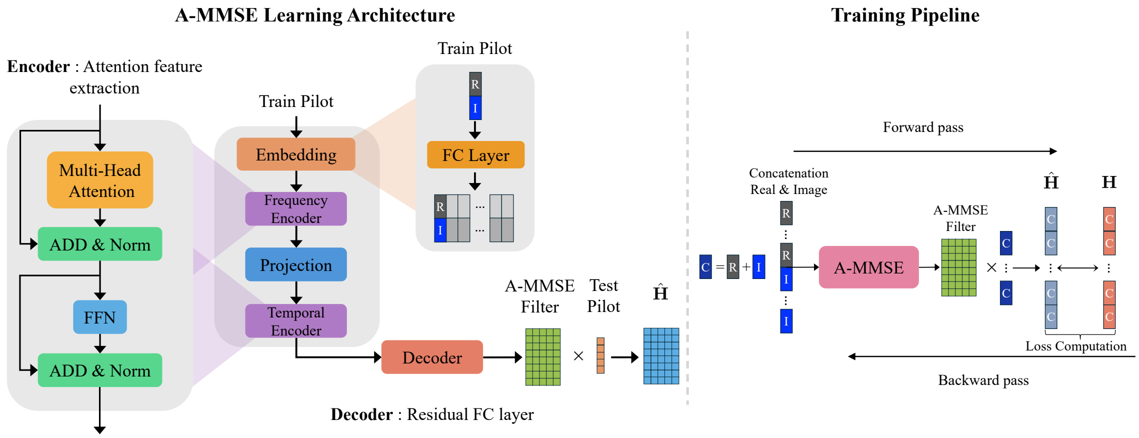


Fig. 1. Overall architecture and end-to-end training scheme of the proposed A-MMSE network.

compared to the oracle MMSE filter $\mathbf{W}^*$. This penalty, or MSE gap [10], is theoretically quantified as:

$$\Delta_{mis} = \mathrm{MSE}(\mathbf{W}_{mis}) - \mathrm{MSE}(\mathbf{W}^*) \tag{5}$$

$$= ||(\mathbf{W}_{mis} - \mathbf{W}^*)\mathbf{\Sigma}_{yy}^{1/2}||_F^2 \geq 0 \tag{6}$$

where $\mathbf{\Sigma}_{yy}$ is the covariance of the received signal. This gap Δ_{mis} establishes a fundamental performance floor that SP-based methods cannot overcome without perfect knowledge.

Second, while data-driven DNN approaches (e.g., ChannelNet, Channelformer) circumvent the need for explicit statistical modeling, they typically operate as end-to-end estimators. This architecture necessitates propagating inputs through deep networks with numerous nonlinear operations (e.g., convolutions, self-attention) during inference. Such computational complexity and lack of flexibility create a bottleneck for real-time processing. These shortcomings motivate the need for a hybrid framework that synergizes the computational efficiency of linear SP filters with the data-driven learning capability of DNNs—precisely the objective of our proposed A-MMSE.

IV. Proposed Attention-aided MMSE

The proposed A-MMSE 1 bridges the gap between linear filter efficiency and deep learning adaptability by transitioning from Neural Estimation to Neural Filtering. Instead of directly mapping pilots to channel estimates, our network generates optimal linear filter coefficients based on the channel context. This architecture decouples non-linear learning from real-time estimation. Consequently, the actual channel estimation relies on efficient linear matrix-vector multiplication, avoiding the high inference latency of conventional end-to-end neural networks.

A. Linear Estimation Mechanism

Conventional deep learning-based approaches typically function as *Neural Estimators*. In this paradigm, the neural network acts as a direct "black-box" estimator that maps noisy pilot observations to channel estimates through a deep sequence of non-linear operations. While effective, this end-to-end strategy inherently couples the learning capability with the inference process, forcing the system to perform computationally expensive non-linear calculations for every estimation instance. In contrast, we propose a paradigm shift towards *Neural Filtering*. The core philosophy of the proposed A-MMSE is to use the deep neural network not as the estimator itself, but as a generator of the optimal estimator. Specifically, the A-MMSE network $\mathcal{F}_{\mathrm{A-MMSE}}$ with parameters θ learns to generate the linear filter matrix $\mathbf{W}_{\mathrm{A-MMSE}}$ based on the complex-valued pilot observations $\mathbf{Y}_p$, split into real and imaginary components:

$$\mathbf{W}_{\mathrm{A-MMSE}} = \mathcal{F}_{\mathrm{A-MMSE}}([\Re(\mathbf{Y}_p); \Im(\mathbf{Y}_p)] \mid \theta) \tag{7}$$

This equation represents the Neural phase, where the network analyzes the pilot context to construct A-MMSE filter elements.

Once the filter matrix is generated, the actual channel estimation is performed solely through a linear operation.

$$\mathrm{vec}\left(\hat{\mathbf{H}}_{\mathrm{A-MMSE}}\right) = \mathbf{W}_{\mathrm{A-MMSE}}\mathbf{Y}_p. \tag{8}$$

This linear neural filtering architecture ensures that the heavy computational burden of deep learning is confined to the filter generation phase (7), while the real-time inference phase relies on the high efficiency of linear matrix-vector multiplication (8). Consequently, A-MMSE successfully bridges the gap between the adaptability of data-driven learning and the structural efficiency of linear filtering.

B. Two-Stage Attention Encoder Architecture

To generate the optimal filter matrix $\mathbf{W}_{\mathrm{A-MMSE}}$, the neural network must effectively capture the statistical correlation structure of the OFDM channel. Under the assumption of a wide-sense stationary uncorrelated scattering (WSSUS) environment, the channel covariance matrix $\mathbf{R}_{\mathrm{full}}$ exhibits a separable structure, approximable as the Kronecker product of frequency-domain correlation $\mathbf{R}_f$ and temporal-domain correlation $\mathbf{R}_t$ (i.e., $\mathbf{R}_{\mathrm{full}} \approx \mathbf{R}_f \otimes \mathbf{R}_t$) [9].

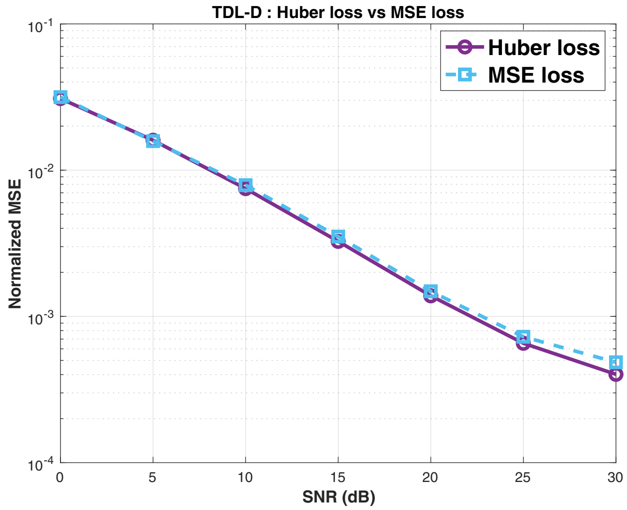


Fig. 2. TDL-D: Channel estimation NMSE comparing Huber loss vs. MSE loss

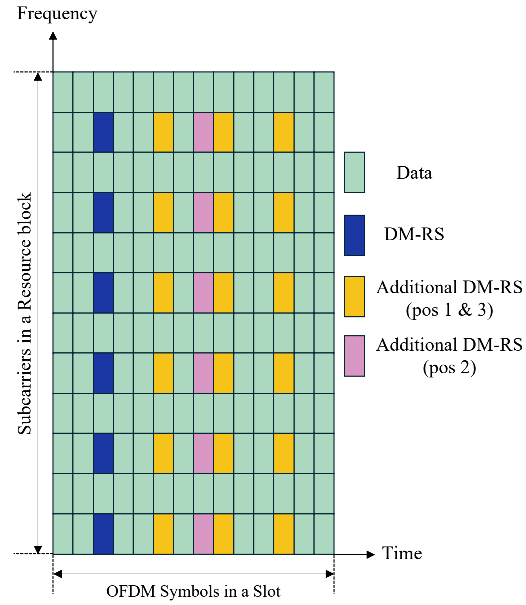


Fig. 3. Structure of a 5G NR resource block with Type-A DM-RS allocation.

Based on this theoretical insight, we design a two-Stage attention Encoder that sequentially learns these decoupled correlation structures.

1) **Stage1: Frequency Encoder**
The first stage focuses on extracting frequency-domain statistical features.
   - **Embedding**: The pilot observations are first linearly embedded into a high-dimensional space with dimension $d_e = N$, where N is the number of subcarriers. This ensures that the embedding explicitly represents the full frequency grid.
   - **Multi-Head Self Attention**: We employ a Multi-Head Self-Attention (MHA) mechanism [11]. Crucially, the number of attention heads is fixed to L_{sym} (the number of pilots per OFDM symbol). This design choice allows each head to attend to specific subcarrier dependencies, effectively modeling the frequency correlation matrix $\mathbf{R}_f$
   - **Frequency Features**: The output of this stage, we call *Frequency features*, denoted as $\mathbf{Y}_{\text{Freq}}$, encapsulates the frequency-domain correlation features inherent in the pilot signals.

2) **Stage2: Temporal Encoder**
The second stage extends the learned frequency features into the temporal domain to model inter-symbol correlations.
   - **Projection**: The frequency features $\mathbf{Y}_{\text{Freq}}$ are projected via a linear layer to a dimension $d_p = NM$, matching the total number of resource elements in an OFDM slot. This projection is essential to map the features into the joint time-frequency space.
   - **Multi-Head Self Attention**: The projected features are processed by a second MHA module. Here, the number of attention heads is set to M (the total number of OFDM symbols). This allows the encoder to capture temporal variations and Doppler effects across symbols, corresponding to the temporal correlation matrix $\mathbf{R}_t$
   - **Full Channel Features**: The final encoded output $\mathbf{Y}_{\text{full}}$ represents the comprehensive joint frequency-temporal statistics of the channel.

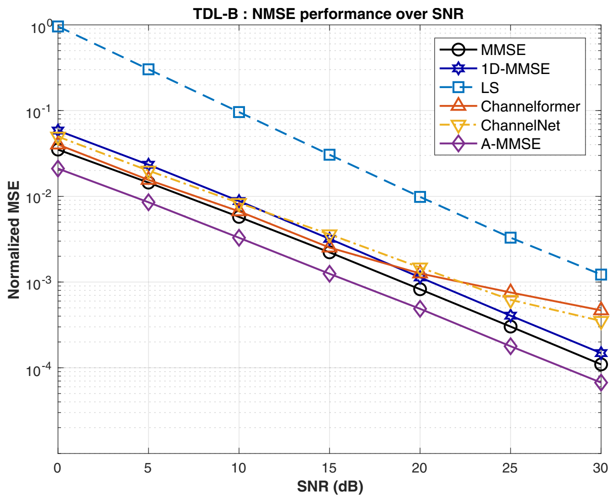
3) **Residual Fully Connected Decoder**
To transform the features $\mathbf{Y}_{\text{full}}$ into the A-MMSE filter matrix, we employ a Residual Fully Connected (ResFC) decoder. The decoder maps the encoded statistical information to the complex-valued coefficients of the A-MMSE filter $\mathbf{W}_{\text{A-MMSE}} \in \mathbb{C}^{NM \times L}$. The output is split into real and imaginary components during generation and recombined to form the final complex filter.

C. Training Strategy

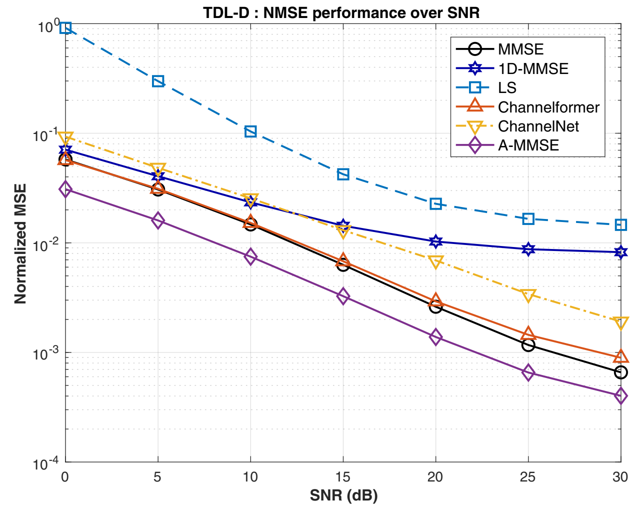
The A-MMSE framework is trained in an end-to-end manner. A key challenge in training is the sensitivity to outliers, which can arise from noise spikes or deep fades. To ensure robust filter learning, we adopt the Huber loss function [12] instead of the standard Mean Squared Error (MSE). The Huber loss $\mathcal{L}_\delta$ is defined as:

$$L_\delta(a_i) = \begin{cases} \frac{1}{2}a_i^2, & |a_i| \le \delta, \\ \delta\left(|a_i| - \frac{1}{2}\delta\right), & |a_i| > \delta, \end{cases} \tag{9}$$

where a_i represents the residual between the true channel and the estimated channel $(\mathbf{H} - \hat{\mathbf{H}})$, and δ is a tuning parameter. Unlike MSE, which grows quadratically and penalizes outliers heavily, Huber loss transitions to a linear penalty for large errors (residuals $> \delta$). This property prevents occasional large noise realizations from destabilizing the gradient descent process, allowing the network to focus on learning the underlying



(a) NMSE performance under the TDL-B Scenario (UMi Street-canyon)



(b) NMSE performance under the TDL-D Scenario (RMa O2I)

Fig. 4. NMSE comparison over SNR: (a) TDL-B and (b) TDL-D scenarios.

channel structure rather than overfitting to noise. Our empirical results as shown in Fig. 2 indicate that Huber loss achieves superior generalization, particularly in high SNR regimes.

V. Simulation Results

A. Simulation Setup

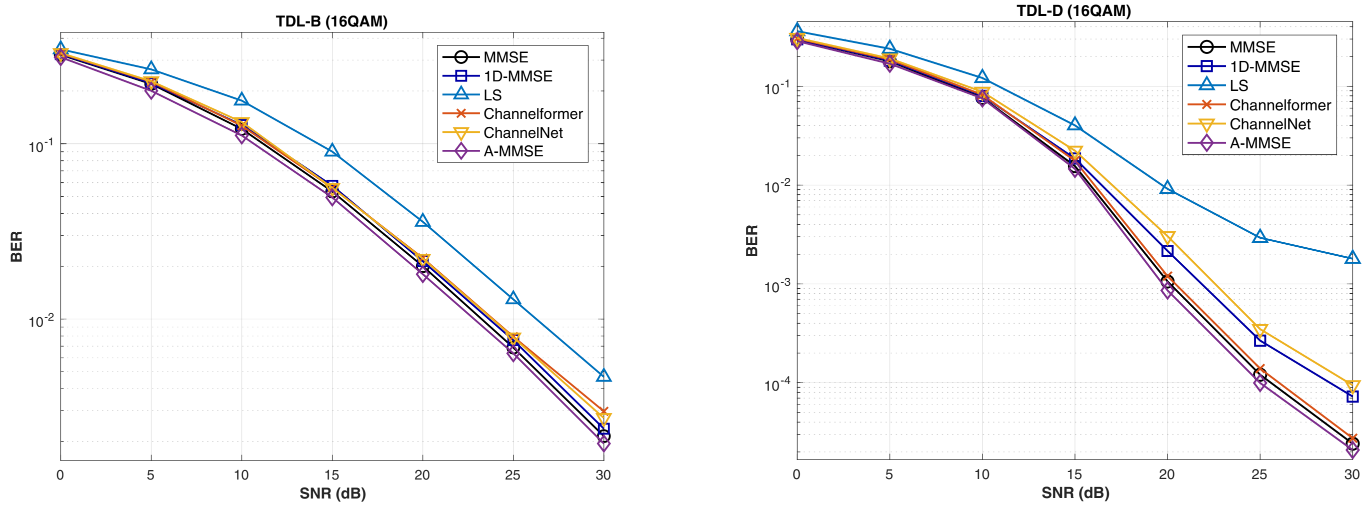
We evaluate the proposed A-MMSE using two representative 3GPP TDL channel models [13] : TDL-B (Urban Micro, 3.5 GHz) representing a typical street-canyon environment, and TDL-D (High Mobility, 5 GHz) modeling a fast-fading scenario with speeds up to 300 km/h. For the pilot configuration, we adopt the 5G NR Type-A Demodulation Reference Signal (DM-RS) [14], [15]. As illustrated in Fig. 3 , we use a comb-2 frequency pattern with an additional time-domain position to track time-varying channel dynamics. This results in DM-RS occupancy on two OFDM symbols (indices 2 and 11) within a standard 14-symbol slot. To capture non-stationary channel statistics, we generated a continuous sequence of 44,000 frames. Crucially, by dynamically updating the channel conditions over time without resets, we intentionally created a non-stationary environment where the wide-sense stationary uncorrelated scattering (WSSUS) assumption is broken globally. This setup specifically reproduces the covariance mismatch scenario discussed in Section III-C, where the discrepancy between the true time-varying statistics and the fixed assumptions leads to an irreducible MSE gap. The dataset was partitioned into 36,000 training, 4,000 validation, and 4,000 testing frames. The model training was conducted using TensorFlow on a single NVIDIA GeForce RTX 4090 GPU.

B. NMSE Performance Analysis

We evaluate the Normalized MSE (NMSE) performance of the proposed A-MMSE compared with baseline schemes across an SNR range of 0 to 30 dB, as illustrated in Fig. 4. In the TDL-B scenario shown in Fig. 4(a), representing a typical urban environment, the A-MMSE yields lower estimation errors than the baselines. At 15 dB SNR, the A-MMSE achieves an NMSE of 1.25×10^{-3}, which is approximately 44% lower than the conventional MMSE (2.22×10^{-3}). Relative to Channelformer (2.52×10^{-3}), the proposed method reduces the NMSE by approximately 50%, suggesting that the Two-Stage Attention Encoder captures the underlying frequency correlations. In the high-mobility TDL-D scenario depicted in Fig. 4(b), the LMMSE exhibits an error floor of approximately 8.2×10^{-3} at high SNR due to channel time-variations, while A-MMSE reaches 4.0×10^{-4} at 30 dB. At 20 dB, the A-MMSE achieves an NMSE of 1.38×10^{-3}, corresponding to a 47% reduction relative to MMSE (2.61×10^{-3}). Additionally, compared with Channelformer (2.92×10^{-3}), the A-MMSE shows a 53% lower NMSE. These results indicate that the Neural Filtering approach addresses non-stationary statistics in dynamic environments.

C. BER Performance Analysis

To evaluate the impact of channel estimation accuracy on symbol detection, we examine the Bit Error Rate (BER) performance for 16-QAM modulation under TDL-B and TDL-D scenarios, as shown in Fig. 5. Similar to the NMSE trends, the BER curves in the low SNR regime (0–10 dB) are dominated by additive noise, resulting in comparable performance across all methods. However, in the high SNR regime, where channel estimation error becomes the limiting factor, the advantages of the proposed method become evident.In the TDL-B scenario as shown in Fig. 5(a), the A-MMSE achieves a BER of 1.95×10^{-3} at 30 dB SNR. This represents a reduction of approximately 9.3% compared with the conventional MMSE (2.15×10^{-3}) and 34.5% relative to Channelformer (2.97×10^{-3}), confirming that improved frequency correlation modeling translates to reliable detection even in



(a) NMSE performance under the TDL-B Scenario (UMi Street-canyon)

(b) NMSE performance under the TDL-D Scenario (RMa O2I)

Fig. 5. NMSE comparison over SNR: (a) TDL-B and (b) TDL-D scenarios.

static environments. The performance gain is more distinct in the high-mobility TDL-D scenario as shown in Fig. 5(b). At 25 dB SNR, the A-MMSE yields a BER of 9.96×10^{-5}, whereas MMSE and Channelformer result in 1.21×10^{-4} and 1.39×10^{-4}, respectively. These figures correspond to BER reductions of approximately 17.5% and 28.5%, respectively. This indicates that the Neural Filtering mechanism effectively mitigates the impact of Doppler-induced aging, maintaining robust link performance in fast-fading conditions.

VI. Conclusion

In this paper, we proposed the Attention-aided MMSE (A-MMSE) framework, a novel approach that transitions from conventional neural estimation to a Neural Filtering paradigm for OFDM channel estimation. By employing a Two-Stage Attention Encoder, our method effectively captures the decoupled frequency and temporal correlation structures of the channel to generate adaptive linear filter coefficients. This design enables the inference phase to be executed via computationally efficient linear matrix-vector multiplication, thereby addressing the high complexity issues inherent in end-to-end neural estimators while bypassing the statistical mismatch limitations of classical MMSE. Simulation results using 3GPP TDL channel models confirm that A-MMSE provides improved NMSE and BER performance compared to existing SP-based and DNN-based baselines, particularly in non-stationary environments. These findings suggest that the proposed architecture offers a favorable balance between estimation accuracy and computational efficiency for practical deployment in next-generation wireless systems .

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